Lab 04
CASCADED ADDERS

Name:	ID:	Section:

4.1. Objectives

After performing this lab, student should be able to:
- Implement full adder using half adders
- Cascade full adders to implement 2-bit adder
- Utilize 7 segment display and driver to see results of binary addition.

4.2. Equipment/Apparatus

1. Adjustable Direct Current (DC) Power supply.
2. Digital Multi Meter (DMM).
3. XOR gate IC 7486.
4. AND gate IC 7408.
5. Light Emitting Diode (LED)
6. Switches.
7. Resistors (330Ω)
8. 7-Segment Display (Common Anode)
9. 7- Segment Driver IC 7447

4.3. Full Adders

The most basic arithmetic operation is the addition of two binary digits. There are four possible elementary operations, namely:

$$0 + 0 = 0$$
$$0 + 1 = 1$$
$$1 + 0 = 1$$
$$1 + 1 = 10)_2 \text{ with 1 Carry}$$

The first three operations produce a sum, whose length is one digit, but when the last operation is performed the sum is two digits. The higher significant bit of this result is called a carry and lower significant bit is called the sum.

A combinational circuit which performs the arithmetic sum of three input bits is called **full adder**. The three input bits include two significant bits and a previous carry bit. A full adder circuit can be implemented with two half adders and one OR gate (figure 4.1). Equation for a full adder can be expressed

$$Sum = A \oplus B \oplus Cin$$

$$Carry = (A \oplus B) \cdot Cin + A \cdot B$$

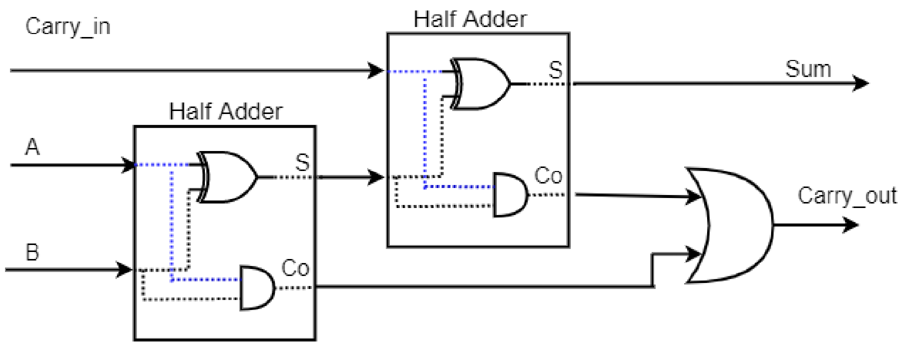

Figure 4.1 - Full Adder created from 2 Half Adder and an OR Gate

Task a)

Construct on bread board two half adders and combine them according to the Logic diagram provided in Figure 4.1.

i. Procedures and observations

1. Implement the logic diagram on the breadboard with necessary power, input and outputs.
    a. Connect inputs A, B and Cin to DIP switches.
    b. Connect output Sum and Carry to LEDs with current limiting resistor.
    c. Setup power supply to provide 5-V.
2. Fix the ICs that are required on the half separator line of breadboard, so there are no shorts.
3. Flip the switches On / Off and check the output for all 8 possible combinations of inputs A, B and Cin.
4. Connect the wire to the main voltage source (Vcc = 5 V) whose another end is connected to last pin of the IC (14th place from the notch).
5. Connect the ground of IC (7th place from the notch) to the ground terminal provided by DC power supply.
6. Switch on the power supply.
7. If the LED light up then output is true, if it stays dark then output is false, which is numerically denoted as logic 1 and logic 0 respectively.
8. Complete the observation Table 4.1.

Table 4.1 – Data Table

C_{in}	A_0	B_0	C_{Out}	S_0
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		

1	1	1		

4.4. Cascading Full Adders

Multiple full adders can be cascaded to add N-bit numbers. Each full adder inputs a C_{in}, which is the C_{out} of the previous adder. Let A_1A_0 and B_1B_0 be two 2-bit binary numbers. The sum of A and B can be denoted as S_1S_0, and carry generated as C_1. The equations can be modified as:

$$S_0 = A_0 \oplus B_0 \oplus C_{in}$$

$$C_0 = (A_0 \oplus B_0) \cdot C_{in} + A_0 \cdot B_0$$

The Carry generated from Least Significant bit (LSB) will serve as C_{in} for next bit addition as shown in Figure 4.

$$S_1 = A_1 \oplus B_1 \oplus C_o$$

$$C_1 = (A_1 \oplus B_1) \cdot C_o + A_1 \cdot B_1$$

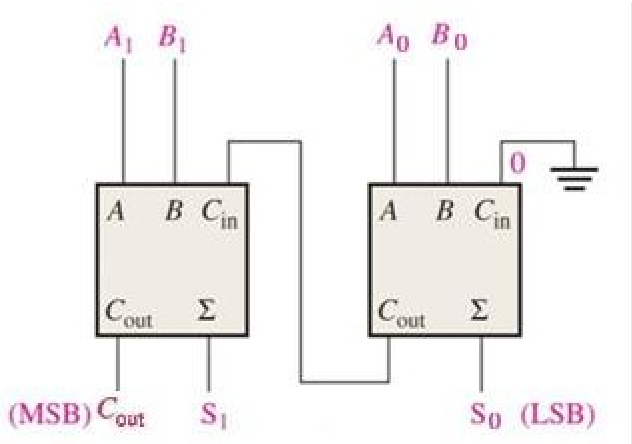

Figure 4.2 - Block diagram

Task b)

Implement the logic diagram provided in Figure 4.2**.** on the breadboard. Two full adders are cascaded where Carry-in of first full stage adder is grounded and Carry$_{out}$ is connected to the Carry-in of second stage Full Adder.

While working complex digital circuits break your circuit into small modules. Implement that module, verify the desired result and then move ahead.

ii. Procedures and observations

1. Connect Inputs (A_1, A_0, B_1, B_0) to slide switches. And Outputs (C_{out}, S_1, S_0) to LEDs.
2. Work with another group to combine two full adders on two breadboards to build two-bit adder.
3. Make necessary changes in the previous adders as per Figure 4.2.
4. Make sure single power supply is used to supply ground and Vcc (5 V) to both bread boards.

5. Flip the switches on / off and check the output for all possible combinations of inputs.
6. Switch on the power supply.
7. If the LED light up then output is true, if it stays dark then output is false, which is numerically denoted as logic 1 and logic 0 respectively.
8. Complete the observation Table 4.2.

Table 4.2 – Data Table

A_1	A_0	B_1	B_0	C_{Out}	S_1	S_0
0	0	0	0			
0	0	0	1			
0	0	1	0			
0	0	1	1			
0	1	0	0			
0	1	0	1			
0	1	1	0			
0	1	1	1			
1	0	0	0			
1	0	0	1			
1	0	1	0			
1	0	1	1			
1	1	0	0			
1	1	0	1			
1	1	1	0			
1	1	1	1			

4.5. Output to Seven Segment display

4.5.1 BCD Numbers

Binary Coded Decimal (**BCD**) is a class of binary encodings of decimal numbers where each digit is represented by a fixed number of bits, usually four. With four bits in binary, we can represent sixteen numbers (0000 to 1111). But in BCD code only first ten of these are used (0000 to 1001). The remaining six code combinations i.e., 1010 to 1111 are invalid in BCD. The decimal weight of each decimal digit to the left increases by a factor of 10. In the BCD number system, the binary weight of each digit increases by a factor of 2. For example, $2^3 = 8$, $2^2 = 4$, $2^1 = 2$ and $2^0 = 1$. The main advantage of the BCD code is that it allows for easy

conversion between decimal and binary numbers. With the 4511 – BCD to 7 Segment Driver we can drive the 7-segment display. Each 4511 IC takes a binary number as an input, then outputs the necessary lines to display that number on the 7-segment display. The relationship between decimal numbers and weighted binary coded decimal digits is given in Table 4.3.

Table 4.3

Decimal Number	BCD 8421 Code
0	0000 0000
1	0000 0001
2	0000 0010
3	0000 0011
4	0000 0100
5	0000 0101
6	0000 0110
7	0000 0111
8	0000 1000
9	0000 1001
10 (1+0)	0001 0000
11 (1+1)	0001 0001
12 (1+2)	0001 0010
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20 (2+0)	0010 0000
21 (2+1)	0010 0001
22 (2+2)	10 0

4.5.2 Seven Segment display

A seven -segment display got its name from the very fact that it has got seven illuminating segments. Each of these segments has an LED (Light Emitting Diode), hence the lighting. The LEDs are so fabricated that lighting of each LED is contained to its own segment. The important thing to notice here is that the LEDs in any seven-segment display are arranged in Common Anode (CA) mode or Common Cathode (CC) mode (common negative). In CA, anodes (i.e., positive) are connected as one terminal whereas in CC all cathodes are connected as one terminal as shown in Figure 4.3.

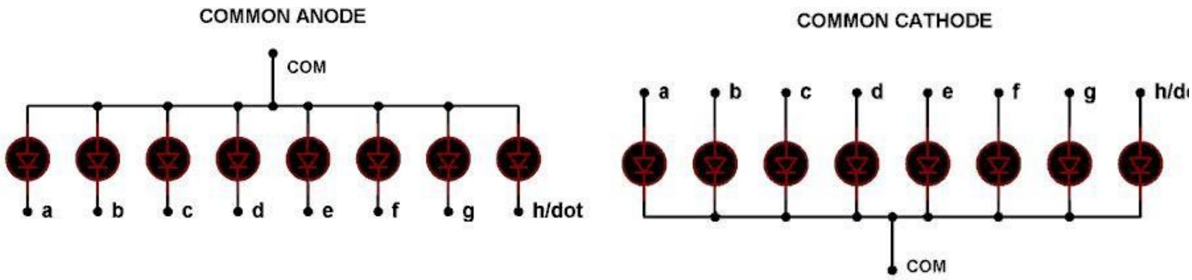

Figure 4.3 - Internal connection of seven segment display

The circuit connection of LEDs in common cathode and common anode is shown in Figure 4.3. Here one can observe that, in CC the negative terminals of every LED is connected and brought out as GND. In CA the positive of every LED is connected and brought out as VCC. These CC and CA come in very handy while multiplexing several cells together. Figure 4.4 shows internal arrangement of LEDs inside a common anode 7-segment.

Tip: Practically one can check each segment of display by using multimeter in diode mode.

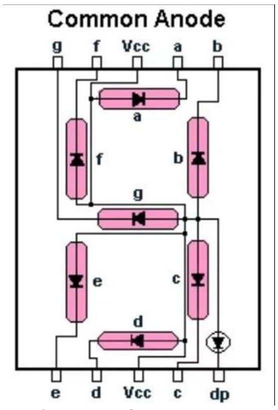

Figure 4.4 - Pin-out of seven segment display.

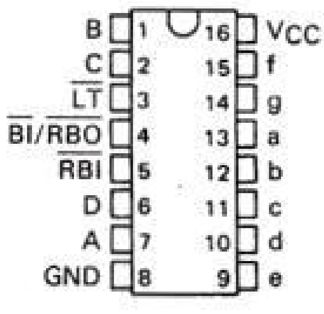

Figure 4.5 - Pin-out of 7447 IC.

4.5.3. Schematic

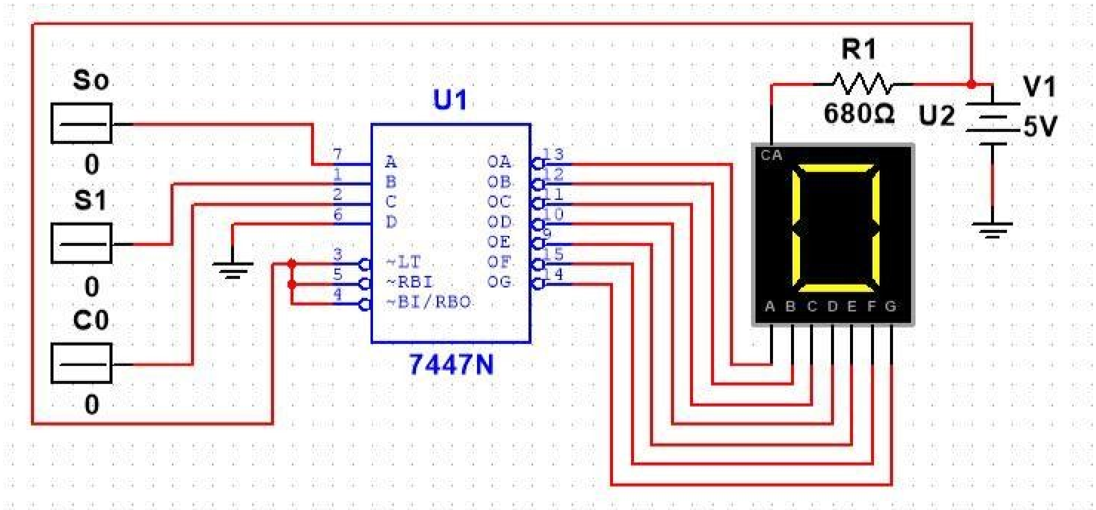

Figure 4.6 -Schematic

Task c)

Implement the logic diagram provided in Figure 4.6 on the breadboard.

i) Procedure and observations

1. Turn off the power supply.
2. Place the IC 7447 on breadboard and make connections as shown in Figure .
3. Connect the wire to the main voltage source (V_{cc} = 5 V) whose other end is connected to last pin of the ICs (16th place from the notch).
4. Connect the ground of ICs (8th place from the notch) to the ground terminal provided by DC power supply.
5. Connect the output of 3 slide switches to pin 2, 1 and 7, the represent the C_{out}, S_1, S_0 respectively of your BCD data.
6. Pin 6 correspond to the MSB of your 4-bit data. Since the data from you full adder will not exceed 3-bits, connect pin 6 of IC to ground.
7. Make the connections for 7 segment display.
8. Turn the power on and observe the results on 7- segment display by flipping the switches.
9. Once the results on your 7-segment display are verified remove the slide switches from your 7447 circuit and connect the output of your 2-bit adder to the 7447 inputs.
10. Verify the result of 2-bit adder on 7-segment display.

Get the results checked by your RA.

ii) Q. How many 7-Segment displays will be required to display the result of 8-bit adders? Justify.

Assessment Rubric
Lab 04
Cascaded Adders

Name:	Student ID:

Marks Distribution:

	Task No.	LR1 Circuit Layout	LR4 Data Collection	LR5 Results	LR11 Report	LR3 Troubleshooting
In - Lab	**Task a**	/10	/05	/05	-	/15
	Task b	/10	/05	/10	-	
	Task c (i)	/10	-	/10	-	
	Task c (ii)				/05	
Total: 100		/30	/10	/25	/05	/15
CLO Mapping		CLO1	CLO1	CLO1	CLO 1	CLO1

Affective Domain Rubric		Points	CLO Mapped
AR 7	Report Submission and Viva	/10	CLO 4
AR 1	Clean up	/05	

CLO	Total Points	Points Obtained
1	85	
4	15	
Total	**100**	

For description of different levels of the mapped rubrics, please refer to the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

Lab Evaluation Assessment Rubrics

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3: Good Points 3	Level 4: Exemplary Points 4
LR1	**Circuit Layout**	Connections between circuit components are mostly wrong. Circuit layout is cluttered. Needs guidance to make correct equipment/ component connections.	Few of the connections made between circuit components are wrong. Circuit layout is not neat and clean as per standards.	Circuit layout and connections are correct but not all connections are neat and clean as per standards.	Neat, clean and correct connections of all the circuit components/ equipment are made as per standard circuit diagram.
LR3	**Troubleshooting**	Unable to identify the fault/minimal effort shown in troubleshooting.	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and takes necessary steps and actions to correct it.
LR5	**Results & Plots**	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR7	**Viva**	Response shows a complete lack of understanding of the assigned / completed task	Response shows shallow understanding of the assigned task	Response shows substantial understanding of the assigned task	Response shows complete understanding of the completed task. The student is able to explain all the related concepts.
LR10	**Analysis**	Results are not interpreted or are analyzed incorrectly.	Analysis of the obtained results is incomplete. Conclusions are not drawn.	Results are analyzed and concluded but are not well explained and justification is not provided with the help of reasoning.	Results are well analyzed supported with reasons. Good comparison is made between the theoretical and experimental results with correct and useful conclusion.
LR11	**Design**	Proposed design is unsubstantiated or does not satisfy most	Justification for proposed design is weak, and it only	Proposed design is substantiated, preferably through	Proposed design is substantiated, preferably through

		of given constraints. The breakdown of system into features is incomplete and still at lower resolution.	satisfies some constraints. The breakdown of system is missing some features and resolution is not appropriate at some places.	proper analysis, and satisfies most given constraints. The breakdown of system into features seems complete, but resolution is not appropriate at some places.	proper analysis, and satisfies all given constraints. The breakdown of system into features seems complete and at appropriate resolution, given students' level.